

Applications of Algebra (Part III) Homework

Name
Date
Course

Write the abbreviated forms of the logarithms below.

① $\log_e 326$

② $\log_{10}(5X)$

③ $\log_e\left(\frac{\pi}{2}\right)$

④ $\log_{10} 2$

Solve the following equations. Write the Roman numeral of each exponent rule that you use in the margin. Round to the nearest thousandth if necessary.

⑤ $e^{3x} \cdot e^{5x} = 6$

⑥ $\frac{10^{9x}}{10^{6x}} = 23$

⑦ $5e^x + 1 = 16$

⑧ $13 = 2(10^x) - 8$

Solve the following equations. Write the Roman numeral of each exponent rule or logarithm property that you use in the margin. Round to the nearest thousandth if necessary.

⑨ $-3 = \frac{\log X}{2} - 4$

⑩ $9 \ln X + 2 = 5$

⑪ $\log 9 + \log X = 0$

⑫ $\ln X - \ln 6 = 1$

⑬ $5 \log 2 - \log X = 2$

⑭ $2 \ln X + \ln 4 = -1$

Each number below represents the hydrogen ion concentration of a liquid. Find the pH of each liquid.

⑮ $[H^+] = 0.00007$

⑯ $[H^+] = 0.03$

Each number below represents the pH of a liquid. Find the hydrogen ion concentration of each liquid.

⑰ $pH = 4$

⑱ $pH = 12$

The following numbers are the lengths of disturbances on a seismometer 100 kilometers from the epicenters of earthquakes. Find the magnitude of each earthquake.

⑲ 0.05 millimeters

⑳ 3,000 millimeters

㉑ 82 millimeters

Find the lengths of disturbances on a seismometer 100 kilometers from the epicenters of various earthquakes if the numbers below represent magnitudes of each earthquake.

②② $M = 2.9$

②③ $M = 8.1$

Answers

① $\ln 326$

② $\log(5x)$

③ $\ln\left(\frac{\pi}{2}\right)$

④ $\log 2$

⑤ $x = \frac{\ln 6}{8} \approx 0.224$

⑥ $x = \frac{\log 2^3}{3} \approx 0.454$

⑦ $x = \ln 3 \approx 1.099$

⑧ $x = \log 10.5 \approx 1.021$

⑨ $x = 100$

⑩ $x = \sqrt[3]{e} \approx 1.396$

⑪ $x = \frac{1}{9} \approx 0.111$

⑫ $x = 6e \approx 16.310$

⑬ $x = \frac{8}{25}$

⑭ $x = \sqrt{\frac{1}{4e}} \approx 0.303$

⑮ $\text{pH} \approx 4.155$

⑯ $\text{pH} \approx 1.523$

⑰ $[\text{H}^+] = 0.0001 \text{ moles/liter}$

⑱ $[\text{H}^+] = 0.000000000001 \text{ moles/liter}$

⑲ $M \approx 1.699$

⑳ $M \approx 6.477$

㉑ $M \approx 4.914$

㉒ $0.794 \text{ millimeters}$

㉓ $125,892 \text{ millimeters}$